

**MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4**

# Mathematics Olympiad Worksheet 08

Level 1 — Beginner · Logical Reasoning II

|             |                                                                       |               |                                                                |
|-------------|-----------------------------------------------------------------------|---------------|----------------------------------------------------------------|
| Class       | <b>Class 4 (Primary · Stage 2)</b>                                    | Subject       | <b>Mathematics</b>                                             |
| Level       | <b>Level 1 — Beginner</b>                                             | Topic Focus   | <b>Logical Reasoning II — Coding, Ranking &amp; Directions</b> |
| Questions   | <b>20</b>                                                             | Time Allowed  | <b>30 minutes</b>                                              |
| Total Marks | <b>20</b>                                                             | Worksheet No. | <b>08 of 100</b>                                               |
| Curriculum  | <b>CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned</b> |               |                                                                |

## About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

## About This Worksheet

Worksheet 08 trains three classic Olympiad reasoning skills. Coding-decoding (letters as numbers, shifting letters, reversing words), ranking (position from front and back, who is tallest or first), and direction sense (left and right turns, distances from the start). These appear in every SOF IMO Logical Reasoning section, and they reward careful, step-by-step thinking.

## Learning Objectives

- Decode letter-number codes and one-step letter shifts.
- Find positions from the front, back, top and bottom of a line.
- Order people by height or finishing position from clues.
- Track facing direction after left and right turns.
- Work out distances after walking forward and back.

## Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



## SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** If **A = 1, B = 2, C = 3** ... then what is the value of **E**?
- (A) 4 (B) 5  
(C) 6 (D) 7
- Q2.** Using **A = 1, B = 2, C = 3** ..., what is the sum of the values of the letters in **CAB**?
- (A) 5 (B) 6  
(C) 7 (D) 8
- Q3.** If **SUN** is coded as **TVO** (each letter moves one step forward), how is **MAP** coded?
- (A) NBQ (B) NBP  
(C) LBQ (D) NCQ
- Q4.** If **DOG** is coded as **COF** (each letter moves one step backward), how is **CAT** coded?
- (A) BZS (B) DBU  
(C) BAS (D) BZT
- Q5.** Five children stand in a line. Anu is 2nd from the front. How many children are BEHIND her?
- (A) 2 (B) 3  
(C) 4 (D) 5
- Q6.** In a queue of 12 people, Ram is 5th from the front. What is his position from the BACK?
- (A) 7th (B) 8th  
(C) 6th (D) 9th
- Q7.** Sita is taller than Gita. Gita is taller than Rita. Who is the SHORTEST?
- (A) Sita (B) Gita  
(C) Rita (D) Cannot say
- Q8.** Ravi faces **North** and turns to his RIGHT. Which direction is he facing now?
- (A) South (B) East  
(C) West (D) North
- Q9.** Meena faces **South** and turns to her LEFT. Which direction is she facing now?
- (A) East (B) West  
(C) North (D) South
- Q10.** A boy walks 5 m towards East, then turns around and walks 3 m back towards West. How far is he from his starting point?
- (A) 8 m (B) 5 m  
(C) 3 m (D) 2 m

**Q11.** If **2 = B**, **4 = D**, **6 = F**, then **8 = ?**

- |       |       |
|-------|-------|
| (A) G | (B) H |
| (C) I | (D) J |

**Q12.** In a race, Karan finished **AHEAD** of Veer but **BEHIND** Aryan. Who came first among the three?

- |           |              |
|-----------|--------------|
| (A) Karan | (B) Veer     |
| (C) Aryan | (D) All tied |

**Q13.** If **TOY** is written as **YOT**, how will **PEN** be written in the same code?

- |         |         |
|---------|---------|
| (A) NEP | (B) ENP |
| (C) PNE | (D) NPE |

**Q14.** A ladder has 10 steps. A cat sits on the 4th step from the bottom. Which step is it from the TOP?

- |         |         |
|---------|---------|
| (A) 6th | (B) 7th |
| (C) 5th | (D) 8th |

**Q15.** The sun rises in the East. In the morning, Riya's shadow falls **BEHIND** her while she walks towards the sun. Which direction is she walking?

- |           |           |
|-----------|-----------|
| (A) East  | (B) West  |
| (C) North | (D) South |

**Q16.** If in a code, **RED** is written as **123** and **DEN** is written as **324**, what number stands for the letter **E**?

- |       |       |
|-------|-------|
| (A) 1 | (B) 2 |
| (C) 3 | (D) 4 |

**Q17.** Four friends sit in a row. Piya is to the immediate **RIGHT** of Tara. Tara is to the immediate right of Nia. Who is to the immediate **LEFT** of Piya?

- |          |                |
|----------|----------------|
| (A) Nia  | (B) Tara       |
| (C) Piya | (D) Cannot say |

**Q18.** Aman's house is 4 km North of the school. The park is 4 km South of the school. How far is Aman's house from the park?

- |          |          |
|----------|----------|
| (A) 0 km | (B) 4 km |
| (C) 8 km | (D) 2 km |

**Q19.** In the word **SMILE**, which letter is exactly in the **MIDDLE**?

- |       |       |
|-------|-------|
| (A) M | (B) I |
| (C) L | (D) S |

**Q20.** If yesterday was Tuesday, what day will **TOMORROW** be?

- |               |              |
|---------------|--------------|
| (A) Wednesday | (B) Thursday |
| (C) Friday    | (D) Monday   |



## ANSWER KEY &amp; DETAILED SOLUTIONS

## Quick Answer Key

| Q   | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8        | 9        | 10       |
|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Ans | <b>B</b> | <b>B</b> | <b>A</b> | <b>A</b> | <b>B</b> | <b>B</b> | <b>C</b> | <b>B</b> | <b>A</b> | <b>D</b> |
| Q   | 11       | 12       | 13       | 14       | 15       | 16       | 17       | 18       | 19       | 20       |
| Ans | <b>B</b> | <b>C</b> | <b>A</b> | <b>B</b> | <b>A</b> | <b>B</b> | <b>B</b> | <b>C</b> | <b>B</b> | <b>B</b> |

## Step-by-Step Solutions

**Q1. Correct answer: (B)**Count: A(1), B(2), C(3), D(4), **E(5)**.**Q2. Correct answer: (B)**C = 3, A = 1, B = 2. Sum = 3 + 1 + 2 = **6**.**Q3. Correct answer: (A)**Move each letter forward by 1: M→N, A→B, P→Q. So MAP → **NBQ**.**Q4. Correct answer: (A)**Move each letter backward by 1: C→B, A→Z (Z comes before A in a loop), T→S. So CAT → **BZS**.**Q5. Correct answer: (B)**Total 5 – Anu's position 2 = **3** children behind her.**Q6. Correct answer: (B)**Position from back = (12 – 5) + 1 = **8th**. Add 1 because Ram himself is counted.**Q7. Correct answer: (C)**Order: Sita > Gita > Rita. The shortest is **Rita**.**Q8. Correct answer: (B)**From North, a right turn faces you **East**. (Remember the clock order: N → E → S → W.)**Q9. Correct answer: (A)**From South, a left turn faces you **East**. (Left from South moves opposite to the clock: S → E.)**Q10. Correct answer: (D)**He went 5 m forward and came 3 m back: 5 – 3 = **2 m** from the start.**Q11. Correct answer: (B)**Each number maps to the letter at that position: 8th letter = **H** (A1 B2 C3 D4 E5 F6 G7 H8).**Q12. Correct answer: (C)**Order: Aryan ahead of Karan, Karan ahead of Veer. So **Aryan** came first.**Q13. Correct answer: (A)**The word is simply written BACKWARDS: PEN → **NEP**.**Q14. Correct answer: (B)**From the top = (10 – 4) + 1 = **7th** step.**Q15. Correct answer: (A)**Walking towards the rising sun means walking **East** (her shadow points West, behind her).

**Q16. Correct answer: (B)**

In RED→123: R=1, E=2, D=3. Check with DEN→324: D=3 ✓, E=2 ✓, N=4. So E = **2**.

**Q17. Correct answer: (B)**

The order from left is: Nia, Tara, Piya, ... So the person immediately left of Piya is **Tara**.

**Q18. Correct answer: (C)**

House is 4 km on one side, park 4 km on the other side of the school:  $4 + 4 = \mathbf{8 \text{ km}}$  apart.

**Q19. Correct answer: (B)**

SMILE has 5 letters: S(1), M(2), **I(3)**, L(4), E(5). The 3rd letter, I, is in the middle.

**Q20. Correct answer: (B)**

Yesterday Tuesday → today is Wednesday → tomorrow is **Thursday**.

